Investigating the Field Around a Wire

Exam code: 0625 & 0972

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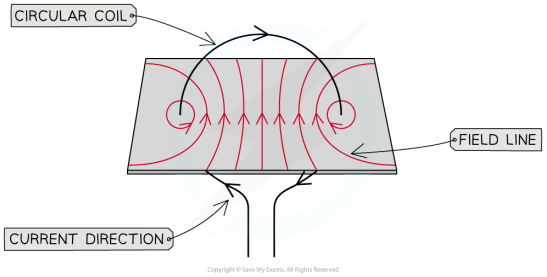
Updated on 30 September 2024

Investigating the magnetic field around a wire

- The magnetic field patterns due to currents in straight wires and in solenoids can be investigated using:
  - a thick wire
  - a solenoid (a wire wrapped into a coil) - for example, a metal slinky
  - cell, ammeter, variable resistor and connecting wires
  - cardboard with holes (the holes must be large enough for the wire to fit through)
  - clamp stand
  - iron filings or a compass
- Spread the iron filings uniformly on the cardboard and place the magnetic needle on the board
- Tap the cardboard slightly and observe the orientation of iron filings
- When the current direction is reversed, the compasses point in the opposite direction showing that the direction of the field reverses when the current reverses



Experiment 1: plotting the magnetic field around a wire

1. Attach the thick wire through a hole in the middle of the cardboard and secure it to the clamp stand
   - Secure the wire vertically so it sits perpendicularly to the cardboard
2. Attach the ends of the wire to a series circuit containing the variable resistor and ammeter on either side of the cell

Help me understand

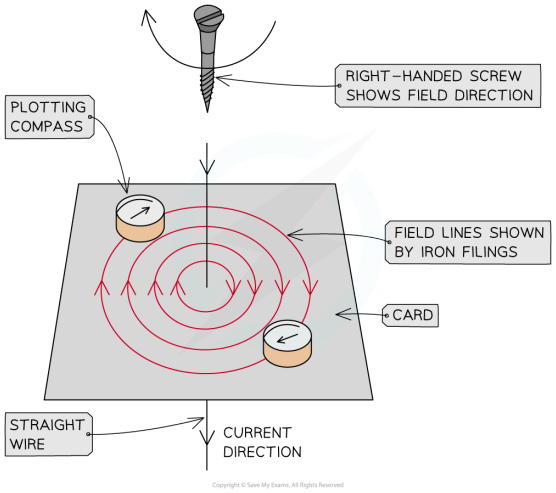
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

each end of the needle once it settles
    - Make sure to draw an arrow to show the direction of the field at different points

2. Move the compass so that it points away from the new dot, and repeat the process above
3. Keep repeating the previous process until there is a chain of dots on the card
4. Then remove the compass, or compasses, and link the dots using a smooth curve – this will be the magnetic field line
5. Repeat the whole process several times to create several other magnetic field lines

Using iron filings:

1. If using iron filings, simply pour the filings onto the cards and gently shake the card until the filings settle in the pattern of the magnetic field around the wire



Experiment 2: plotting the magnetic field around a solenoid

1. Attach the thick wire through a hole on one side of the cardboard and loop it through a hole on the other side of the cardboard and secure it to the clamp stand
    - Secure the wire so it forms a circular loop around the cardboard

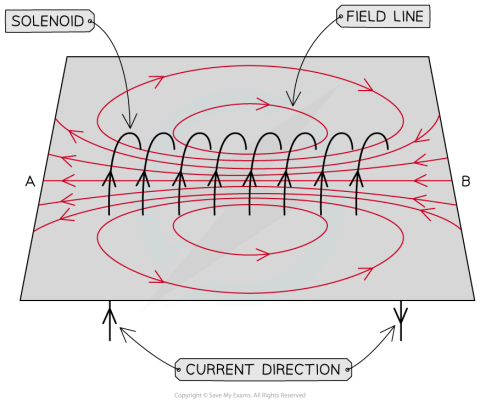
2. Attach the ends of the wire to a series circuit containing the variable resistor and ammeter on either side of the cell

Using plotting compasses:

1. Follow the procedure outlined in Experiment 1
    - Note: this can be carried out using a solenoid, but since a solenoid is essentially many circular loops, the pattern around a circular loop can be extended to give the pattern around a solenoid

Using iron filings:

1. Take a solenoid (a metal slinky works well for this) and thread it through pre-made holes in a piece of card



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